

AI Grounding / IQ

Choose trusted context for one customer decision

Use this deck to discuss sources, access, retrieval evidence, and proof before a pilot ships.

Workshop outcome: choose a governed context pattern, access boundary, evaluation plan, and operating evidence for one bounded AI-assisted decision.

How to use this conversation

Choose the smallest useful scenario that can be grounded, evaluated, and operated safely. This is not an enterprise platform design session.

Each module moves through the same three steps:

Step	Customer question
Discuss	What decision are we improving, and what constraints matter?
Decide	Which path fits the source, access model, and experience?
Prove	What must be true before we move on?

Record the choice, owner, open question, and evidence gate as you go. Keep implementation details in the scenario modules.

Module 1 context: provision the grounding foundation

Agree on the foundation before content enters the pilot.

Customer discussion prompts:

- Which business decision or action needs trusted context?
- Which environment will host the pilot, and who owns it?
- Which region, identity model, logging boundary, and data handling expectations apply?
- Which model and retrieval capabilities must be available in that environment?

Principle: protect the source of truth before optimizing developer convenience.

Module 1 choices: shared foundation or existing landing zone?

Choice	When it fits	Trade-off
Scenario-provisioned foundation	Fast pilot with clear defaults	Customer still needs operational ownership
Existing Azure landing zone	Enterprise controls are already in place	More dependency on customer platform teams
Portal-led setup	Early orientation or stakeholder demo	Harder to reproduce without a written contract
Bring-your-own resources	Customer already has Foundry, search, and monitoring	Must prove configuration matches the pilot needs

Discuss identity, network boundaries, observability, and regional capacity before ingestion.

Module 1 — what must be true: foundation is ready for content

Move forward when the team can show:

- A named environment owner and service owner
- Keyless-first access plan for people and workload identities
- Model and retrieval capabilities available in the chosen region
- Monitoring and trace destination named before pilot traffic
- Environment values documented as a reusable contract, without secrets

Decision: it is safe to connect approved pilot content.

Module 2 context: select the source and permission architecture

Start grounding with source ownership, not a search box.

Ask the customer:

- Which system remains the source of truth?
- Who owns content quality, retention, and retirement?
- Which users may see which records, and why?
- Is permission trimming needed at retrieval time, answer time, or both?
- What happens when content is missing, conflicting, or restricted?

Non-negotiable: every pilot source needs an owner and an access decision.

Module 2 choices: match IQ and source patterns to the work

Pattern	Best fit	Watch for
Copilot Studio + SharePoint	M365 workflow over governed SharePoint content	Limited need for custom orchestration
Foundry IQ	Custom app or agent needing managed grounding over approved knowledge	Confirm source support, region, and permissions
Azure AI Search	Custom indexing, scoring, or retrieval control	You own more ingestion and access behavior
Fabric IQ	Analytical facts, measures, and business semantics	Keep metrics aligned to governed data products
Work IQ	Microsoft 365 collaboration and work context	Respect user-scoped permissions and privacy
Web IQ	Curated public information with attribution	Treat as external context, not the system of record

Module 2 — what must be true: signed source and access decision

The source decision is ready when it records:

- Approved source or sources, with business owner and steward
- Source-of-truth statement and freshness expectation
- Permission model, including restricted and access-denied cases
- Citation, provenance, and audit expectations
- Boundary between governed internal knowledge, live work data, analytics, and public information

Decision: the pilot has a defensible source architecture before indexing starts.

Module 3 context: ingest and index approved content

Ingestion does more than move files. It keeps the evidence behind an answer.

Discuss:

- Which approved content is small enough for the pilot but representative enough to test?
- What metadata must survive ingestion: source, version, owner, date, sensitivity, access group?
- How will stale, duplicate, superseded, and conflicting documents be handled?
- Which content should stay remote rather than copied?

Goal: retrieve the right passage and provenance for the user.

Module 3 choices: managed, custom, or remote knowledge

Choice	Use when	Trade-off
Foundry IQ managed ingestion	The platform-supported source pattern fits	Less custom control, more managed behavior
Azure AI Search indexer	Source can be pulled on a schedule	Indexer limits shape metadata and enrichment options
Push indexing	Custom chunking, enrichment, or source pipeline is required	Team owns more code and operational failure modes
Content preprocessing	Documents need extraction, layout, or structure first	Adds another quality gate before retrieval
Remote knowledge source	Content should remain in place	Availability and permissions depend on the remote system

Do not copy content just to make retrieval look simpler.

Module 3 — what must be true: approved documents are discoverable

Before moving on, reviewers should see:

- Representative pilot corpus indexed or connected
- Required source metadata visible in retrieval results
- Permission-sensitive examples behaving as designed
- Golden questions returning plausible source passages
- Known stale or conflicting content labeled with the agreed behavior

Decision: the knowledge layer produces trustworthy retrieval evidence before you judge answers.

Module 4 context: compare chat and embedding choices

Model choice matters. Judge it against the grounding task.

Discuss what the pilot needs from:

- The chat or query-planning model: reasoning, instruction following, latency, cost, region, and safety
- The embedding model: retrieval quality for the customer's vocabulary, abbreviations, and document style
- Capacity: expected users, peaks, and whether throughput should be reserved or consumption-based

Principle: pick models using golden-dataset evidence, not reputation.

Module 4 choices: quality, latency, cost, and capacity

Decision	Customer trade-off
Larger chat model	May improve reasoning, but can increase latency and cost
Smaller chat model	May be faster and cheaper, but needs proof on hard cases
Embedding candidate	Must improve retrieval for this corpus, not just benchmark well
Regional deployment	Keeps data and latency aligned, but may constrain availability
Pay-as-you-go	Flexible for pilots, with variable throughput
Provisioned throughput	Predictable capacity, with stronger planning commitment

Access control and citations are baseline obligations, not trade-offs.

Module 4 — what must be true: model decision supports grounded answers

A model choice is ready when the team has compared:

- Retrieval hit quality on golden questions
- Answer usefulness with citations and abstention behavior
- Latency and cost under expected pilot usage
- Region and quota feasibility
- Failure behavior on ambiguous, stale, or restricted questions

Decision: the selected model and embedding path meet the pilot's quality and operating constraints.

Module 5 context: build retrieval before adding an agent

A grounded experience should prove retrieval before it adds orchestration.

Customer discussion prompts:

- Can the system find the right source passage for the user's question?
- Does the answer cite the source and avoid unsupported claims?
- What should happen when evidence is weak, missing, stale, or access denied?
- Which user action is safe after the answer is shown?

Design stance: retrieval, citations, abstention, and recency come before agent behavior.

Module 5 choices: answer synthesis or extractive retrieval?

Choice	When it fits	Trade-off
Knowledge-base answer synthesis	Managed grounding can produce useful cited answers	Less direct control over answer construction
Extractive retrieval plus custom prompt	Team needs stronger control over tone and policy	More prompt and evaluation ownership
Direct hybrid query	App needs transparent retrieval and custom ranking	More engineering effort before user value

State the required behaviors: cite available evidence and abstain when it is missing. Do not reveal whether missing evidence is absent or restricted.

Module 5 — what must be true: retrieval is trustworthy enough to use

The grounded app is ready for the next step when:

- Golden questions return relevant sources with provenance
- Answers quote or summarize only supported evidence
- Access-denied cases do not leak restricted content
- Stale-source cases trigger the agreed warning or refusal
- Reviewers can explain why an answer was accepted or rejected

Decision: add an agent only when orchestration solves a customer need that retrieval alone cannot.

Module 6 context: add agent and live-data routing only when justified

Use agents and tools when the experience needs planning, source routing, or action boundaries. They cannot fix unclear knowledge architecture.

Discuss:

- Which requests should use policy knowledge, analytics, work context, public information, or live operational data?
- Which source is authoritative for each request type?
- Which live-data calls are safe, necessary, and auditable?
- What must the agent refuse or route away from?

Boundary: live data complements the source of truth. It must not silently replace it.

Module 6 choices: route sources with clear rules

Choice	Use when	Trade-off
Foundry agent with knowledge tool	The experience needs orchestration over approved knowledge	Requires explicit instructions and routing tests
Multi-source knowledge routing	Several governed knowledge sets answer different intents	Source boundaries must be visible and testable
Live-data tool	A real-time status, inventory, case, or transaction check is required	Tool reliability, permissions, and audit enter the pilot scope
Multi-agent workflow	Distinct roles need separate responsibilities	More moving parts and harder evaluation

Choose the simplest route that preserves source authority and user trust.

Module 6 — what must be true: routing is explainable and controlled

Routing is ready when the team can show:

- Test questions for each source boundary
- Correct route chosen for policy, analytics, work-context, web, and live-data cases
- Refusal or escalation for unsupported requests
- No live-data call without a justified real-time need
- Trace or log evidence that shows which source answered

Decision: evaluate the agent as a controlled router.

Module 7 context: evaluate and trace

Decide whether the assistant is ready for real people.

Discuss:

- Which golden questions represent normal, edge, stale, restricted, and adversarial cases?
- Who reviews correctness, citations, access behavior, and business usefulness?
- What traces are needed to investigate a failure without exposing unnecessary sensitive content?
- Who signs off that the evidence is enough?

Operating mindset: an evaluation you only read is a report. One that blocks release is a control.

Module 7 choices: how much evidence is enough

Decision	Customer trade-off
Strict evaluation gate	Slower release, stronger trust for high-impact decisions
Phased pilot gate	Faster learning, with narrower scope and clear limits
Trace detail	Better troubleshooting, balanced against privacy and retention
Continuous evaluation	Better drift detection, with ongoing reviewer commitment
Manual review	Higher judgment quality, but limited scale
Automated checks	Better repeatability, but must be calibrated against human review

Generic quality metrics miss correct refusal, current citation, and silence about restricted content.

Module 7 — what must be true: the assistant is good enough

Evidence to produce:

- Golden-set results for correctness, citation, abstention, and access behavior
- Adversarial cases, including prompt injection hidden in retrieved content, with the mitigation re-tested
- One request traced end to end, so a wrong answer can be explained rather than guessed at
- The permission probe re-run against the agent

Decision: the evidence supports release to real people, or identifies what must change first.

Module 8 context: deploy and surface it to users

Decide where people use the assistant and who runs it.

Discuss:

- Where does this audience already work: Teams, an existing app, or somewhere new?
- Who is in the pilot, and how is access granted and revoked?
- Who triages a wrong answer, and how does a user report one?
- What ends the pilot?

Delivery check: connect the answer path you tested, with or without an agent. Prove it through the intended user's authenticated channel.

Module 8 choices: pick the doorway, keep the rules

Surface	Where users meet it	When it wins
Your own app or API	An existing front end	Pilots with one consumer
Publish to Teams and M365 Copilot	Teams, Copilot app	Users already work there; no new app to adopt
Copilot Studio	Teams, Copilot app	Only if the source decision was SharePoint and M365
Foundry hosted agent	A dedicated endpoint	Other teams consume it, or you need runtime control
Custom web UI	A purpose-built app	Demo, custom auth, or a required response contract

Five rules stay fixed: no keys; preserve the permission boundary; carry tracing into the runtime; pin the version; roll back by repointing.

Module 8 — what must be true: safe to hand to real users

The release contract records:

- The surface, why it was chosen, and who can use it
- A pinned agent version and a rollback measured in minutes
- The permission probe re-run against the surface itself, not just the agent
- A named triage owner, a review cadence, and how users report a bad answer
- A pilot exit criterion with a date, and a signed release decision

Decision: ship, ship with conditions, or stop. A written stop decision has value.

Close the discussion: turn decisions into the pilot plan

Bring the people who can approve sources, permissions, evaluation, and operations.

Before the next working session, record:

- the bounded customer decision and non-goals
- the source of truth, source owner, and access boundary
- the selected IQ or source pattern and any live-data boundary
- the golden dataset, including restricted and stale-source cases
- the evidence gate for retrieval, routing, evaluation, tracing, and release
- one owner and due date for every unresolved decision

Exit outcome: a pilot decision record the delivery team can implement through the modules.